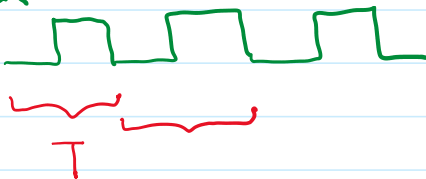


05 12
 06h 18
 16
 32

duty cycle:

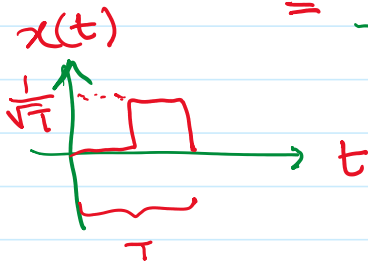
$$f = 12 \text{ MHz} \rightarrow T = \frac{1}{f} = \frac{1}{12 \times 10^6} = 0.08 \times 10^{-6}$$

clock



duty cycle := Time for which the pulse is high
 T

$$= \frac{T/2}{T} = \frac{1}{2} = 0.5$$



$$E_x = \int_0^T x^2(t) dt = \int_0^{T/2} (0)^2 dt + \int_{T/2}^T \left(\frac{1}{\sqrt{T}}\right)^2 dt$$

$$= \int_{T/2}^T \frac{1}{T} dt = \frac{1}{T} \int_{T/2}^T dt$$

$$\begin{aligned}
 & T_{1/2} \\
 & = \frac{1}{T} \left\{ t \right\}_{T_{1/2}}^T \\
 & = \frac{1}{T} \left\{ T - \frac{T}{2} \right\} = \frac{1}{T} \left\{ \frac{2T - T}{2} \right\}
 \end{aligned}$$

$$E_n = \frac{1}{2}$$

Buzzer



```

HERE:  JNB  P2.3,
        SETB P1.5
        CLR  P1.5
        SJMP HERE
  
```

Qw2

→ AGAIN JNB P.2, HERE
 ↓
 SJMP AGAIN
 → HERE CLR P1.5
 SETB P1.5
 ...

SETB P1.5

SJMP AGAIN